The Liquidator's Frontier

Leverage, Seniority and the Endogenous Attachment Point
in Isolated Lending Markets

σ-Labs

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Abstract

A collateralised borrow in an isolated lending market is not a loan with a haircut. It is a two-tranche structure: the borrower holds the residual claim and is long a put on the collateral, the lender holds a senior one. What is not standard is that the attachment point is of two kinds. The *contractual* attachment — the price at which senior loss becomes arithmetically unavoidable — is fixed at creation and has a closed form. The *effective* attachment — the price at which the mechanism stops working — is produced block by block by an oracle, a liquidation incentive and an interest-rate model, is never written down, and is the one that binds.

This paper derives the first in closed form for markets of the Morpho Blue type, then spends its length on the second. Two corrections to the received account follow. The borrower's put is struck at the senior threshold, not at the liquidation threshold, so a liquidated borrower is not a wiped-out one; and the naive account, which ignores the premium paid to the agent who enforces the structure, overstates both the equity cushion and the buffer by exactly that premium — 2.9 points of the entry price at production parameters, against a 7.9-point interval between the two thresholds.

The central result concerns not when liquidation is permitted but when it is performed. Modelling a flash-financed liquidator, who receives collateral at the oracle price and must sell it at the market price, gives one incentive inequality. It yields a maximum tolerable slippage σ^* and a minimum viable position size proportional to gas. That budget is not free: σ^* and the room the liquidator has to work in are the same primitive $1 - \lambda$, split in the fixed proportion c to $1 - c$, so what pays the enforcing agent is taken out of the distance over which it must succeed.

The inequality is then made dynamic. Slippage need not be estimated — the venue state is readable at every block — so the frontier is a floor computable each block, free of any distributional assumption, at the cost of being a best case. A position larger than one admissible slice must be cleared in sequence, and we bound how many slices a market can carry by a number independent of the borrower's leverage. Impact, volatility and carry then enter one first-passage problem whose degenerate limits are the deterministic and driftless cases, and two markets with identical caps and identical room are shown to differ by orders of magnitude in the probability of senior loss. Finally we bound where a liquidation cascade stops — with finite borrow it always stops — by a secant condition valid at the cliffs of a step function, where a contraction argument is not.

Three apparently distinct failure modes — a reverting oracle, illiquid collateral, and dust positions nobody bothers to close — are the same inequality losing on different terms. We close with the consequence for parameter choice: raising the loan-to-value cap does not only thin the residual tranche, it shrinks the enforcing agent's slippage budget, so the cap is a claim about the collateral's liquidity stated in the grammar of a risk parameter.

Keywords. Collateralised lending; tranching; attachment point; liquidation incentives; flash loans; market microstructure; decentralised finance; lending protocols.

1 The setting

Fix an isolated lending market in the sense of Morpho Blue: one collateral asset, one loan asset, one oracle, one interest-rate model, and one liquidation loan-to-value cap $\lambda \in (0,1)$, all immutable at creation.

A borrower holds collateral worth A in units of the loan asset, valued at the oracle price, and owes D. Write

$$\mathrm{LTV} \;=\; \frac{D}{A}, \qquad E \;=\; A - D, \qquad L \;=\; \frac{A}{E} \;=\; \frac{1}{1-\mathrm{LTV}},$$

so that $D = (L-1)E$ and $\mathrm{LTV}_0 = (L-1)/L$ at entry. The position is *healthy* while

$$A\lambda \;\geq\; D, \tag{1}$$

which is the on-chain solvency predicate, evaluated with the oracle price and nothing else.

Throughout, ℓ denotes a proportional fall in the price of the collateral, so the collateral is worth $A(1-\ell)$ after the move, and D is unchanged over the horizon considered unless interest is explicitly in play.

1.1 Why this is a tranche

Three positions on the same asset must be distinguished, because they are routinely conflated.

Holding the collateral outright gives a linear payoff with unbounded downside. Posting it and borrowing gives something else: the borrower keeps the drawn D, and if the collateral falls far enough it is seized and the debt is extinguished against it. The borrower's loss is therefore truncated, and the lender's recovery is $\min(D,\, A/\mathrm{LIF})$ for the incentive factor LIF defined below — which is D less $1/\mathrm{LIF}$ puts on the collateral struck at $D\,\mathrm{LIF}$. The lender is short those puts and the borrower is long them, which is what seniority means in credit.

Two remarks on the strike, both of which the informal account gets wrong. It is $D\,\mathrm{LIF}$, the level at which exhaustive seizure stops covering the debt — not the liquidation threshold D/λ, which is a barrier at which the mechanism switches on, not a level at which anyone is paid. And the two are distinct: at the liquidation threshold, exhaustive seizure returns $A - D\,\mathrm{LIF} > 0$ to the borrower. Being liquidated is not being wiped out.

Re-levering does not create the residual tranche; the first borrow already did. It thins it.

2 Tranche geometry

Proposition 1 (Liquidation threshold)**.** *A position entered at* $\mathrm{LTV}_0 \leq \lambda$ *becomes liquidatable after a proportional price fall*

$$\ell_{\mathrm{liq}} \;=\; 1 - \frac{\mathrm{LTV}_0}{\lambda} \;=\; 1 - \frac{L-1}{L\lambda}.$$

Proof. By (1) the position ceases to be healthy when $A(1-\ell)\lambda < D$, that is $1-\ell < D/(A\lambda) = \mathrm{LTV}_0\,/\lambda$. Substituting $\mathrm{LTV}_0 = (L-1)/L$ gives the second form. □

Enforcement is not free. A liquidator who repays r receives collateral worth $r\,\mathrm{LIF}$, where the incentive factor is fixed by two protocol constants.

Definition 2 (Liquidation incentive factor)**.** With cursor c and cap M,

$$\mathrm{LIF}(\lambda) \;=\; \min\!\left(M,\; \frac{1}{1-c\,(1-\lambda)}\right), \qquad c = 0.3, \quad M = 1.15.$$

LIF is decreasing in λ: a tighter cap pays the enforcer more. The reason is a constraint, not a judgement about risk — the bonus has to fit inside the collateral surplus $1/\lambda - 1$ that still exists when liquidation opens, and that surplus shrinks as the cap rises. Throughout we assume M is not binding, which holds for $\lambda > 0.5652$ and therefore for every parameter considered here.

Proposition 3 (Senior threshold and loss)**.** *Exhaustive seizure of collateral worth A repays at most A/LIF. Consequently senior loss is unavoidable if and only if $A < D\,\mathrm{LIF}$, that is after a fall*

$$\ell_{\mathrm{sen}} \;=\; 1 - \mathrm{LTV}_0\ \mathrm{LIF},$$

and its magnitude is $D - A/\mathrm{LIF}$, not $D - A$.

Proof. A liquidator repaying r removes $r\,\mathrm{LIF}$ of collateral value, so the largest repayment the collateral can fund is $r_{\max} = A/\mathrm{LIF}$. If $A/\mathrm{LIF} \geq D$ the debt is fully repayable and $A - D\,\mathrm{LIF}$ of collateral survives. Otherwise the residual debt $D - A/\mathrm{LIF}$ is unrecoverable. Writing $A = A_0(1-\ell)$ and dividing by A_0 gives the threshold. □

The threshold ℓ_{sen} is the *contractual attachment point* of the structure: the fall at which the senior claim begins to be impaired, determined by λ, fixed at creation, and by LTV_0, fixed when the position is opened and thereafter drifting only with carry. It is available in closed form to anyone who reads the parameters. It is the object a static tranching analysis would stop at. Sections 3 to 5 are about the other one.

The correction matters at the boundary. At $A = D$ — the point a naive account would call break-even — the lender has already lost $D(1 - 1/\mathrm{LIF})$, which is 4.20% of the debt at $\lambda = 0.86$ and 11.25% at $\lambda = 0.625$.

Definition 4 (Buffer)**.**

$$B \;=\; \ell_{\mathrm{sen}} - \ell_{\mathrm{liq}} \;=\; \mathrm{LTV}_0\left(\frac{1}{\lambda} - \mathrm{LIF}\right).$$

B is the interval of collateral price over which liquidation is both permitted and sufficient: the room the mechanism has to work in, measured from the entry price.

Proposition 5 (The buffer is never empty)**.** *$B > 0$ for every $\lambda \in (0,1)$, every $\mathrm{LTV}_0 > 0$, every cursor $c < 1$ and every cap M.*

Proof. $B > 0$ iff $\lambda\,\mathrm{LIF} < 1$. Since $\mathrm{LIF} \leq 1/(1 - c(1-\lambda))$ by definition of the minimum, $\lambda\,\mathrm{LIF} \leq \lambda/\big((1-c) + c\lambda\big)$, and $\lambda < (1-c) + c\lambda \iff (1-c)\lambda < 1 - c \iff \lambda < 1$. No case split on M is needed. □

The positivity is a design constraint rather than a discovery: the incentive formula is built to satisfy $\mathrm{LIF} < 1/\lambda$, which is what makes the structure a tranche at all rather than one in which the enforcer is paid out of the lender's principal from the first block.

The buffer can be small. At $L = 3$ and $\lambda = 0.86$, liquidation opens at -22.5%, senior loss begins at -30.4%, and $B = 7.9$ points. The naive account — which asks only when the equity is extinguished, at a fall of $1/L = 33.3\%$ — is precisely the case $\mathrm{LIF} = 1$. It places the liquidation threshold correctly, since ℓ_{liq} does not involve LIF at all, and misses the senior threshold by

$$\tfrac{1}{L} - \ell_{\mathrm{sen}} \;=\; \mathrm{LTV}_0\,(\mathrm{LIF} - 1),$$

the enforcer's premium expressed as a fraction of the entry price — 2.9 points here. The buffer inherits exactly that error, since B is the difference of the two thresholds: the naive counterpart of B is $1/L - \ell_{\mathrm{liq}} = 10.8$ points against a true 7.9. Note that $1/L$ and B are not themselves comparable quantities; the first is a cushion measured from entry, the second an interval between two thresholds.

Two measures of the same room will both be needed below, and confusing them is the easiest error to make here. B is measured from the *entry* price and therefore depends on L. Measured from the price at which liquidation opens — which is where the mechanism actually operates — the distance to the senior threshold is

$$\beta \;=\; 1 - \lambda\,\mathrm{LIF}, \tag{2}$$

which does not depend on L at all. At $\lambda = 0.86$, $\beta = 10.23\%$ while $B = 7.93\%$. Every dynamic statement in this paper is about β.

Proposition 6 (Partial liquidation below ℓ_{sen})**.** *Let $r \in \big(0, \min(D,\, A/\mathrm{LIF})\big)$. After a repayment r the ratio becomes $\mathrm{LTV}' = (D - r)/(A - r\,\mathrm{LIF})$, and*

$$\mathrm{LTV}' > \mathrm{LTV} \iff A < D\,\mathrm{LIF}.$$

Moreover the eventual shortfall $D - A/\mathrm{LIF}$ is invariant under any such repayment.

Proof. $\mathrm{LTV}' > \mathrm{LTV}$ iff $(D - r)A > D(A - r\mathrm{LIF})$, i.e. $-rA > -Dr\,\mathrm{LIF}$; dividing by $r > 0$ and reversing gives $A < D\,\mathrm{LIF}$. For the second claim, $(D - r) - (A - r\mathrm{LIF})/\mathrm{LIF} = D - r - A/\mathrm{LIF} + r = D - A/\mathrm{LIF}$. □

This is the structural content of Proposition 3. Above the senior threshold each partial liquidation lowers the loan-to-value ratio and the position converges to health; below it, each partial liquidation *raises* the ratio and the position never returns to health. The threshold marks where the mechanism stops being able to repair a position, and it does so before any bad debt is recorded.

It does not, however, mark a point at which liquidating destroys value, and the temptation to read it that way should be resisted. The invariance in Proposition 6 says the shortfall is fixed the moment the price crosses ℓ_{sen}: partial liquidation below the threshold neither deepens nor cures it. If anything the incentive runs the other way. Every unit repaid below the threshold converts an exposure that can still worsen into a loss that cannot, so from the lender's side faster clearing is strictly better there — the opposite of what the rising ratio suggests. What changes at ℓ_{sen} is only that the position can now be worked to exhaustion rather than back to health.

3 The liquidator's frontier

Propositions 1–6 describe when liquidation is *permitted*. A lending market is cleared not by permission but by an agent who acts when acting pays.

Consider a liquidator financed by a flash loan, which is the standard construction and imposes no capital requirement. The agent borrows r of the loan asset, repays r of debt, receives collateral worth $r\mathrm{LIF}$ *at the oracle price*, sells it into the market at proportional slippage σ, returns the flash loan with fee ϕ, and pays gas g:

$$\pi \;=\; r\,\mathrm{LIF}\,(1-\sigma) \;-\; r(1+\phi) \;-\; g.$$

Proposition 7 (Incentive inequality)**.** *Liquidation of size r is performed only if*

$$\mathrm{LIF}\,(1-\sigma) \;\geq\; 1+\phi+\frac{g}{r}. \tag{3}$$

On Morpho Blue $\phi = 0$: the flash-loan primitive transfers the assets out and pulls the same amount back, charging nothing. We take $\phi = 0$ below.

Corollary 8 (Slippage tolerance)**.** *In the limit of large r, the market clears only where*

$$\sigma \;\leq\; \sigma^* \;=\; 1-\frac{1}{\mathrm{LIF}}.$$

Corollary 9 (Minimum viable size)**.** *At gas g and slippage $\sigma < \sigma^*$, no liquidation occurs below*

$$r_{\min} \;=\; \frac{g}{\mathrm{LIF}\,(1-\sigma)-1}, \qquad \textit{so at } \sigma = 0, \quad r_{\min} = \frac{g}{\mathrm{LIF}\,\sigma^*}.$$

For $\sigma \geq \sigma^*$ no size is viable.

The second corollary is often quoted as a gas bound. It is not: $r_{\min}$ is proportional to gas but inversely proportional to σ^*, and Table 1 shows it moving twentyfold across the range of caps at fixed g. The cap sets the scale of the dust; gas only sets the unit.

3.1 What the budget costs

σ^* is not a free parameter of the design. It is carved out of the same quantity that supplies the buffer.

Proposition 10 (Partition of the cap headroom)**.** *Away from the cap M,*

$$\sigma^* \;=\; c\,(1-\lambda), \qquad \beta \;=\; (1-c)(1-\lambda)\,\mathrm{LIF},$$

so that the headroom $1-\lambda$ splits as

$$1-\lambda \;=\; \underbrace{c\,(1-\lambda)}_{\textit{enforcer's budget}} \;+\; \underbrace{(1-c)(1-\lambda)}_{\textit{room retained}}, \qquad \textit{and} \qquad \frac{\beta}{\sigma^*} \;=\; \frac{(1-c)\,\mathrm{LIF}}{c}.$$

Proof. $\sigma^* = 1 - 1/\mathrm{LIF} = 1 - \big(1 - c(1-\lambda)\big) = c(1-\lambda)$. For β, use (2): $1 - \lambda\mathrm{LIF} = \big(1 - c(1-\lambda) - \lambda\big)\mathrm{LIF} = (1-c)(1-\lambda)\mathrm{LIF}$. The ratio follows. □

This is the paper's central identity and it is worth stating without symbols. The cursor c is a *split*. Every point of loan-to-value headroom the protocol declines to lend against is divided between the premium that pays the enforcing agent and the room that agent is given to work in, in the proportion c to $1-c$. The retained room is a distance in loan-to-value; converting it to a price distance multiplies by LIF, which is why the ratio the liquidator actually experiences, β/σ^*, is $(1-c)\mathrm{LIF}/c$ — at production constants, roughly one part budget to 2.4 parts room. The two cannot be set independently: a protocol wishing to pay liquidators more for a given cap can do so only by shortening the distance over which they have to succeed. And since $\sigma^* = c(1-\lambda)$ is linear, the exchange rate is a constant: *every point of loan-to-value cap costs the liquidator thirty basis points of slippage budget*, whatever the collateral and whatever the venue.

Remark 11 (The budget is also the loss). σ^* is likewise the lender's proportional loss at $A = D$, since $(D - A/\mathrm{LIF})/D = 1 - 1/\mathrm{LIF}$ there. This is an identity rather than a theorem, but it is the reason the two halves of this paper describe one object: the premium that pays the enforcer, the market depth the collateral must have, and — at the point $A = D$ — the loss the lender absorbs when that depth is absent are the same number.

λ	LIF	σ^*	β	$r_{\min}$ at $g = 50$
0.980	1.0060	0.60%	1.41%	8,283
0.965	1.0106	1.05%	2.48%	4,712
0.945	1.0168	1.65%	3.91%	2,980
0.915	1.0262	2.55%	6.11%	1,911
0.860	1.0438	4.20%	10.23%	1,140
0.770	1.0741	6.90%	17.29%	675
0.625	1.1268	11.25%	29.58%	394

Table 1: The frontier at production parameters. In loan-to-value units the split is the constant $(1-c)/c = 2.33$; as a ratio of price distances $\beta/\sigma^* = (1-c)\mathrm{LIF}/c$ runs from 2.35 to 2.63 across the range. $r_{\min}$ is in loan-asset units and evaluated at $\sigma = 0$; any positive slippage raises it.

Remark 12 (What σ measures). The collateral is transferred at the *oracle* price and realised at the *market* price, so σ is the oracle-to-execution gap and not merely order-book depth. A stale oracle marking the collateral above the bid consumes the same budget as thin liquidity. An oracle that reverts sets $\sigma = \infty$, since the trade cannot be initiated at all: the solvency predicate (1) cannot be evaluated, and the enforcement transaction fails before it reaches the market. Withdrawal of collateral requires the same evaluation and freezes with it; lending is not symmetrically frozen, as §6 records.

4 Clearing

Corollary 8 bounds a rate. A lending market is exposed to a quantity. This section turns one into the other.

4.1 Slippage is observable, and part of its path is deterministic

Fix the argument of σ once, since it is easy to lose. Write $\sigma(v)$ for the proportional slippage incurred selling collateral of loan-asset value v into the venue, with σ increasing and $\sigma(0^+) = 0$. A liquidation repaying r sells $r\,\mathrm{LIF}$, so the frontier (3) reads $\mathrm{LIF}(1-\sigma(r\mathrm{LIF})) \geq 1 + g/r$ throughout.

It is tempting to treat σ as a random variable to be estimated. On an on-chain venue it is not: the state is readable at every block, so σ is a deterministic function of public state at the moment of evaluation. Two qualifications are owed immediately. Liquidators frequently exit through venues that are not on-chain, for which this is false; and the state at the moment the frontier binds is not the state now, since liquidation follows an adverse move.

Part of that second gap is nonetheless deterministic. The debt is denominated in the loan asset and does not fall with the collateral, so extinguishing it after a fall ℓ requires selling $1/(1-\ell)$ times as many units. On a venue whose depth in units is unchanged this alone takes a position from 2% slippage to 3.3% after a 40% fall. The effect is real but its size is model-dependent: on a constant-product venue the reserve of collateral grows as the price falls, impact scales as $1/\sqrt{1-\ell}$ rather than $1/(1-\ell)$, and the same move gives 2.6%. The honest statement is that the requirement grows with the move under any depth profile, and that the growth rate has to be read off the venue rather than assumed.

Remark 13 (The frontier gives a floor, computable each block)*.* A curator can evaluate (3) at each block against a grid of hypothetical ℓ and obtain a threshold: the largest move this market absorbs while remaining enforceable. It should be read as a best case, not a stress test. Current state plus the quantity effect is a *floor* on slippage — it assumes the venue does not withdraw quotes — and therefore an *upper bound* on the tolerable move. In stress, quote withdrawal routinely dominates the quantity effect. The construction's value is that it is exact and requires no distributional assumption; its limit is that the true threshold is always below it.

4.2 How many slices a market can carry

Definition 14 (Admissible slice)**.** The largest repayment a single profitable liquidation can carry, ignoring gas, is

$$q_{\max} \;=\; \sup\{\, q > 0 \;:\; \sigma(q\,\mathrm{LIF}) \leq \sigma^* \,\},$$

which is positive since $\sigma(0^+) = 0 < \sigma^*$, and finite whenever σ is unbounded.

A position of size $Q > q_{\max}$ cannot be cleared in one transaction at a profit. It must be sold in $N = \lceil Q/q_{\max} \rceil$ slices, separated by whatever interval Δ the venue needs for arbitrage to restore the price. Write $T = N\Delta$ for the clearing time. Let $\rho \in [0,1]$ denote the fraction of each slice's impact which is permanent, that is, not recovered between slices.

The liquidator therefore pushes the price toward the very threshold it is racing against. Two opposing effects have to be tracked, and the first version of this argument tracked only one. Each slice moves the price down by $\rho\sigma^*$, which consumes room. But each slice also repays debt, and since seizure removes collateral value LIF times faster than it

removes debt, the *proportional* distance from the current price to the senior threshold widens as the position is worked. The bound below is stated conservatively, discarding the second effect.

Because impacts compound rather than add, the natural unit here is the log-decline. Write

$$\kappa \;=\; -\ln(1-\rho\sigma^*), \qquad b \;=\; -\ln(\lambda\mathrm{LIF}), \tag{4}$$

for the permanent log-impact of one slice and the log-distance from the liquidation price to the senior threshold. To first order $\kappa \simeq \rho\sigma^*$ and $b \simeq \beta$; both are used in this form for the rest of the paper.

Proposition 15 (Slice-count bound)**.** *Suppose clearing begins at the liquidation threshold, the price is unchanged except by the liquidator's own permanent impact, and the oracle tracks that impact between slices. Then the senior tranche survives the clearing of a position spanning N slices whenever*

$$N \;<\; \frac{b}{\kappa}, \qquad \textit{to first order} \qquad N \;\le\; N^* \;=\; \frac{\beta}{\rho\,\sigma^*} \;=\; \frac{(1-c)\,\mathrm{LIF}}{c\,\rho}.$$

The bound is independent of the borrower's leverage and, up to LIF*, of the cap.*

Proof. Let $x_n = A_n/\mathrm{LIF}$ be the repayment capacity of the collateral after n slices and D_n the debt, so senior loss is the event $x_n < D_n$. A slice repaying q seizes collateral worth qLIF and then moves the price by a permanent factor $(1-u)$, $u = \rho\sigma^*$, so $x_{n+1} = (x_n - q)(1-u)$ and $D_{n+1} = D_n - q$. Unrolling with equal slices $q = D_0/N$,

$$x_N \;=\; x_0(1-u)^N - q\sum_{k=1}^{N}(1-u)^k, \qquad D_N \;=\; D_0 - Nq,$$

so the slack $S_N = x_N - D_N$ satisfies

$$S_N \;=\; x_0(1-u)^N - D_0 + q\Big[N - \sum_{k=1}^{N}(1-u)^k\Big] \;\ge\; x_0(1-u)^N - D_0,$$

the bracket being non-negative since $(1-u)^k \le 1$. Enforcement begins at $A_0 = D_0/\lambda$, so $x_0 = D_0/(\lambda\mathrm{LIF})$ and $S_N \ge x_0\big((1-u)^N - \lambda\mathrm{LIF}\big)$, which is positive exactly when $N\ln(1-u) > \ln(\lambda\mathrm{LIF})$, that is $N < b/\kappa$. Replacing $1-(1-u)^N$ by Nu gives the first-order form, and Proposition 10 the closed form. □

The oracle hypothesis is not cosmetic and it cuts both ways. If the oracle tracks the impact, the damage appears where the proof puts it: as movement of A_n toward the senior threshold. If the oracle lags, A_n is marked too high, the position looks safer than it is, and the impact surfaces instead as slippage on the next slice — shrinking $q_{\max}$, raising N, and tightening the same bound from the other side. A lagging oracle does not avoid the constraint; it relocates it from the numerator to the denominator and conceals it while it accumulates.

At $\rho = 0.3$ the bound is $b/\kappa = 8.51$, or $N^* = 8.12$ in first-order form: *a position more than about eight admissible slices in size cannot be assumed to survive its own unwinding, even in a market that never moves.* The first-order constant is the partition ratio of

Proposition 10 divided by ρ, which is why it does not depend on how levered the borrower was — a fact the entry-based buffer B obscures.

The bound is sufficient, and slack. Under the same hypotheses, restoring the deleveraging effect discarded in the proof — the bracket dropped there — and iterating to a full clear in equal slices gives the sharp threshold: 17 slices at $\rho = 0.3$, 26 at $\rho = 0.2$. The gap is slack in the bound, not uncertainty about the venue, and it decomposes cleanly: $8.12 \to 8.51$ is the compounding restored by the logarithm, $8.51 \to 17$ is the deleveraging. The threshold does, however, depend on how the position is worked, and in a direction worth stating. Under the proposition's own model — impact $\rho\sigma^*$ per slice whatever the slice's size — splitting a slice in two doubles the damage, and the sharp threshold falls from 17 to 8 at half-slices and 4 at quarter-slices. The coarsest admissible schedule is therefore optimal, and 17 is the best case over schedules rather than the only case. The conclusion survives a more realistic impact function so long as it is concave or linear in size: with impact proportional to size the threshold runs from 17 at whole slices to 16.8 in the continuous limit, so slicing moves it by about one per cent, and with square-root impact finer slicing is again worse. Only convex impact would let a liquidator do better by slicing, and that is not the usual shape. What matters is that these numbers are small, and that they are properties of the venue rather than of the loan.

This is the point at which the second attachment can be named.

Definition 16 (Effective attachment point)**.** Write $d_{\text{liq}} = -\ln(1 - \ell_{\text{liq}})$ and $d_{\text{sen}} = -\ln(1 - \ell_{\text{sen}})$, so that $d_{\text{sen}} - d_{\text{liq}} = b$ exactly. The *effective attachment point* of a position of size Q, with $N = \lceil Q/q_{\max} \rceil$, is

$$d_{\text{eff}}(Q) \;=\; \max\bigl(d_{\text{liq}},\; d_{\text{sen}} - N\kappa\bigr), \qquad \ell_{\text{eff}}(Q) \;=\; 1 - e^{-d_{\text{eff}}(Q)},$$

a lower bound on the smallest exogenous fall from which clearing can no longer complete before the senior claim is impaired.

The definition is a rearrangement of the slack inequality in the proof above, not a new object: the position must survive N slices of permanent impact, so the fall it can absorb beforehand is the contractual one less the damage the clearing will do. In log units that subtraction is exact, and the guard against large N is automatic — $d_{\text{sen}} - N\kappa$ simply falls through d_{liq} and is clipped there.

Two properties follow, and neither was available for ℓ_{sen}. First, $\ell_{\text{liq}} \le \ell_{\text{eff}} < \ell_{\text{sen}}$, the floor attained exactly at $N = b/\kappa$ — the two thresholds of §2 are the two ends of this one quantity, and b/κ is where the room has been entirely consumed by the act of clearing. The upper end is approached but not attained, since even a single slice does κ of damage: $\ell_{\text{eff}}(q_{\max}) = 29.8\%$ against $\ell_{\text{sen}} = 30.4\%$ at the parameters below. That last 0.6 points is slack of the bound rather than a property of the position; the sharp recursion, which credits the seizure before the impact, recovers it. Second, ℓ_{eff} depends on Q, on $q_{\max}$ and on ρ, none of which is a parameter of the market and all of which move without any transaction on the protocol; Δ enters too, once the price is allowed to move on its own, through Proposition 17. At $\lambda = 0.86$, $L = 3$ and $\rho = 0.2$, a position of five admissible slices has $\ell_{\text{eff}} = 27.4\%$ against a contractual 30.4%; at ten slices, 24.3%.

This is the sense in which the attachment point of these structures is endogenous. It is not a term of the contract but an outcome of the venue, recomputed every block, and

it is the one that binds. Substituting the sharp recursion for the slack bound gives

$$\ell_{\text{eff}}^{\text{sharp}}(N) \;=\; 1-(1-\ell_{\text{sen}})\,(1-u)\,\frac{(1-u)^{-N}-1}{Nu}, \qquad u=\rho\sigma^{*},$$

which restores $\ell_{\text{eff}}(q_{\max})=\ell_{\text{sen}}$, halves the initial slope in N, and moves the floor from b/κ to 17. What the two forms share is only the concave descent between the two thresholds, once N is measured as a fraction of its own threshold.

4.3 Impact, volatility and carry in one race

Proposition 15 assumes the price moves only through execution. The opposite idealisation — that it moves only through news — is equally available and equally partial. Both are limits of one problem.

Write $Z_t=\ln\!\big(A_t/(D_t\text{LIF})\big)$. Senior loss is the event $Z_t\le 0$, and $Z_0=\ln\!\big(1/(\lambda\text{LIF})\big)=:b$ at the moment enforcement begins. Three things move Z: the liquidator's permanent impact, at rate $m=\kappa/\Delta$ with κ from (4) — Z is a logarithm and one slice multiplies the price by $1-\rho\sigma^{*}$, so κ and not $\rho\sigma^{*}$ is the per-slice drop; the carry, at rate r_b-y where r_b is the borrow rate and y the collateral's yield; and exogenous price innovation, with volatility σ_p. Repayment is omitted from Z, exactly as it is from the bound of Proposition 15: by Proposition 6 each slice raises Z, so the probability below is an upper bound, and its $\sigma_p\to 0$ limit inherits the slack of N^{*} rather than the sharp threshold of 17.

Proposition 17 (Probability of senior loss)**.** *Suppose $Z_t=b-\mu t+\sigma_p W_t$ with $\mu=m+(r_b-y)$, and let clearing take $T=N\Delta$. Then*

$$\mathbb{P}(\textit{senior loss before } T) \;=\; \Phi\!\left(\frac{\mu T-b}{\sigma_p\sqrt{T}}\right) \;+\; e^{2\mu b/\sigma_p^2}\,\Phi\!\left(\frac{-\mu T-b}{\sigma_p\sqrt{T}}\right).$$

Proof. Standard first passage for arithmetic Brownian motion: for $X_t=-\mu t+\sigma_p W_t$ and $a<0$, $\mathbb{P}(\min_{t\le T}X_t\le a)=\Phi\big((a+\mu T)/(\sigma_p\sqrt{T})\big)+e^{-2\mu a/\sigma_p^2}\Phi\big((a-\mu T)/(\sigma_p\sqrt{T})\big)$. Set $a=-b$. □

Corollary 18. *As $\sigma_p\to 0$ the probability tends to $\mathbf{1}\{\mu T>b\}$, the value at $\mu T=b$ being $\frac{1}{2}$; ignoring carry, $\mu T=N\kappa$ and the survival condition is $N<b/\kappa=8.51$, which is exactly Proposition 15. As $\mu\to 0$ it tends to $2\Phi\big(-b/(\sigma_p\sqrt{T})\big)$, the driftless reflection-principle bound.*

Call $(b-\mu T)/(\sigma_p\sqrt{T})$ the *clearing margin*: room remaining after the liquidator's own damage, divided by the distance the price is expected to wander while the position is worked. Its content is that λ, and therefore β, is uninformative alone. Two markets at the same cap, and therefore with identical buffers, can sit at opposite ends of the outcome space.

Remark 19 (Where carry actually bites). In Proposition 17 carry enters μ additively with impact, which places the interest-rate model and the venue's depth in the same units — but over a clearing window it is negligible: at a 300% spread, 2.5 hours of carry is 9 basis points against 840 of impact. Carry is not what loses the race. It is what starts it: at a flat price and $r_b>y$, the position reaches the liquidation threshold at

	deep, moderate vol.	**thin, high vol.**
$q_{\max}$	2,000,000	1,000,000
position Q	10,000,000	10,000,000
slices N	5	10
Δ	5 min	15 min
T	25 min	2.5 h
σ_p	100%/yr	300%/yr
ρ	0.2	0.2
b	10.79%	10.79%
impact μT (carry omitted)	4.22%	8.44%
noise $\sigma_p\sqrt{T}$	0.69%	5.07%
clearing margin	**9.53**	**0.46**
$\mathbb{P}$(senior loss)	$\approx 10^{-21}$	41%

Table 2: Identical cap, identical room b, opposite outcomes. Both positions satisfy Proposition 15 at $\rho = 0.2$, so neither fails on impact alone. The difference is carried entirely by $q_{\max}$, Δ and σ_p, none of which appears among the market's parameters. The clearing margin, not the probability, is the result: the left-hand entry is a Gaussian tail at nine and a half standard deviations and should not be read as a number. Under a Student t with three degrees of freedom the two margins give terminal tails of $\approx 10^{-3}$ and 33%, against Gaussian terminal tails of 7×10^{-22} and 32% — a factor of a few hundred rather than of 10^{21}, and the ordering unchanged. The 41% above includes the reflection term; the terminal tails do not.

$t^* = \ln(\lambda/\,\mathrm{LTV}_0)/(r_b - y)$, and r_b is set by an adaptive model that raises it precisely when utilisation rises — that is, when borrowers cannot repay and lenders cannot withdraw. At $L = 3$ a position with t^* of over a decade at a 2% spread has a t^* of about a month at a 300% one, with nothing about the position changed. The interest-rate model does not decide whether liquidation succeeds; it decides how many positions are queued at the frontier when it is tested.

5 Cascades

Proposition 15 treats one position pushing the price against itself. In a market with many borrowers the same impact makes *other* positions liquidatable, and the question is how far the resulting sequence runs.

Declines must now be added across positions entered at different prices, so we work in logarithms throughout this section, where impact compounds additively and no first-order approximation is needed. Write $d = -\ln(1-\delta)$ for a log-decline measured from the pre-shock price, and recall κ and b from (4). All positions are re-marked at the pre-shock price, so $d_{\mathrm{liq}}(i)$ is position i's liquidation threshold expressed in that common frame.

Let $d_0 > 0$ be an exogenous log-decline and let $Q(d)$ be the borrow rendered liquidatable once the price has fallen by d. Q is a primitive of the market, read off the distribution of liquidation prices across open positions; it is a non-decreasing step function, with a jump

wherever a mass of positions shares a liquidation price. Clearing that volume obeys the mechanics of §4.2 — $Q(d)/q_{\max}$ slices, each leaving κ of permanent log-impact — so

$$d_{n+1} \;=\; F(d_n), \qquad F(d) \;=\; d_0 + \kappa\,\frac{Q(d)}{q_{\max}}, \qquad d_1 = F(d_0). \tag{5}$$

Proposition 20 (Where a cascade stops)**.** *A least $d \geq d_0$ satisfying*

$$Q(d) \;\leq\; \frac{q_{\max}}{\kappa}\,(d - d_0) \tag{6}$$

exists whenever total borrow is finite, with no further hypothesis on Q. If in addition the market holds finitely many positions, say m, the iterates (5) *reach it in at most $m+1$ rounds, and the set of positions liquidated grows strictly at every round but the last.*

Proof. Q is non-decreasing, so F is non-decreasing and maps the interval $I = [d_0,\, d_0 + \kappa Q_{\mathrm{tot}}/q_{\max}]$ into itself, Q_{tot} being total borrow. I is a complete lattice and (6) is $F(d) \leq d$ rearranged, so by Tarski's fixed point theorem the set of fixed points of F in I is a non-empty complete lattice and has a least element, which is the least d satisfying (6). Neither continuity nor finiteness is used.

For the second claim, let $S_k = \{i : d_{\mathrm{liq}}(i) \leq d_k\}$ be the set liquidated after k rounds. The iterates are non-decreasing — $d_1 \geq d_0$ since $Q \geq 0$, and monotonicity of F propagates it — so the S_k are nested increasing. If $S_{k+1} = S_k$ then $Q(d_{k+1}) = Q(d_k)$, since Q depends on d only through that set, hence $d_{k+2} = d_{k+1}$ and the iteration has stopped. So every round but the last adds at least one position, and there are m to add. □

The two halves of the proposition need different hypotheses, and separating them is the point. Existence is a lattice fact and costs nothing. What finiteness buys is that the *iteration* reaches the fixed point, and reaches it fast — a market of a thousand positions settles in at most a thousand and one rounds, and in simulation far fewer. Without finiteness the limit of the iterates need not satisfy (6): a right-continuous Q with infinitely many jumps accumulating at some d^* lets the sequence converge to d^* while $F(d^*) > d^*$, so the least d satisfying (6) lies strictly above the limit. Real markets have finitely many positions and the question does not arise; a continuum model of the same market would need Q left-continuous instead.

Remark 21 (Relation to network clearing). Proposition 20 is, in method, the clearing problem of Eisenberg and Noe [3]. Their clearing vector solves $p = \bar{L} \wedge (e + \Pi^{\top} p)$, with $\bar{L}$ the vector of total liabilities, e the endowments and Π the relative liability matrix. The map is monotone on a lattice, so its solutions form a complete lattice and existence is a lattice fact — Tarski [5], not Banach — and their fictitious default algorithm terminates in at most one round per node, because each round either adds a defaulter or stops. Both arguments are the ones used above, and the second is why the bound is $m+1$ rather than merely "eventually".

The economics is not theirs, and the difference matters. In [3] node i reaches node j through a *contractual liability*: Π is a balance sheet. Borrowers in an isolated market have no claim on one another whatsoever; position i reaches position j only by moving a price, which is fire-sale contagion, and the closer ancestor is Cifuentes, Ferrucci and Shin [1], who layer a price that falls with aggregate sales onto the Eisenberg–Noe problem and obtain a joint fixed point in payments and price. The reduced-form version used by

Greenwood, Landier and Thesmar [4] and the estimates in Duarte and Eisenbach [2] are the empirical descendants.

Two further departures are worth naming. Their state is a vector on a general liability graph; ours collapses to a scalar, because in an isolated market every position reaches every other through exactly one price — the network is a star through the venue, not a graph, and that is a property of the market design rather than a modelling convenience. And their algorithm iterates *downward* from full payment and selects the greatest clearing vector, where (5) iterates *upward* from the shock and selects the least fixed point above d_0. The direction is not a choice: a cascade genuinely begins at the shock and works up, so the least fixed point is the one the market visits.

Two things follow from the form of (6). It is a secant condition, not a derivative one, so it needs no smoothness and is evaluated exactly at the cliffs of Q, which are the only places the question is interesting; a contraction argument would be both unavailable and unnecessary. And it is a statement about *where* the cascade stops, not whether. In a market with finite borrow it always stops. The useful question is never termination — it is whether $\bar{d}$ arrives before the senior thresholds of the positions in its path.

Read (6) against Proposition 15 and the two agree. A single position starting at its own liquidation threshold makes $\bar{d} - d_0 = N\kappa$, and its senior claim is impaired only if that exceeds b — that is, only if $N > b/\kappa = 8.51$, which is Proposition 15's $N^* = 8.12$ up to the compounding the logarithm now carries exactly. The market case is the same inequality with Q read off many borrowers instead of one. The quantity $q_{\max}/\kappa$ is the venue's *marginal depth* in this model: the borrow that may be made liquidatable per unit of log-decline without the decline feeding itself. The condition reads

the borrow made liquidatable over an interval of decline must not exceed the venue's marginal depth times the width of that interval.

Both sides are observable. The left is a property of how borrowers are distributed; the right is a property of the market they will be sold into.

Two consequences deserve statement. First, the condition is *local*: it can hold on average across the price range and fail on an interval. A market in which many positions share a liquidation price has a jump in $Q(\cdot)$ there, and a cascade that reaches that jump is carried past it regardless of how comfortably (6) holds elsewhere. It is only reached if no earlier d satisfies (6) first, which is precisely why the criterion must be evaluated as a secant from d_0 rather than pointwise. Concentration of *liquidation prices*, not merely of size, is the hazard — and a front end offering one default leverage produces it whenever entries also cluster in time, since a common leverage and a common entry price give a common liquidation price.

Second, the condition couples two things that no single agent controls. A curator sets caps and therefore influences Q; nobody sets σ. A market can pass the test on the day it is configured and fail it after liquidity migrates, with no transaction on the protocol having occurred.

Finally, (6) delivers the effective attachment point at market scale. Borrower i can suffer senior loss only if the terminal $\bar{d}$ exceeds $d_{\mathrm{liq}}(i) + b$, which in logarithms is exactly $d_{\mathrm{sen}}(i)$, with no first-order term discarded. The condition is necessary and not sufficient: a position cleared early in the cascade is sold before the price reaches its own senior level. Since $\bar{d}$ is

non-decreasing in d_0, the market-scale effective attachment of position i is well defined as the smallest shock that carries the cascade past it, $d_{\text{eff}}(i) = \inf\{d_0 : \bar{d}(d_0) > d_{\text{sen}}(i)\}$, with $\ell_{\text{eff}}(i) = 1 - e^{-d_{\text{eff}}(i)}$ in the units of §2. When Q has a single step of height Q this is $d_{\text{sen}} - \kappa Q/q_{\max}$, which is Definition 16 exactly. The contractual attachment of Proposition 3 is a property of that position's parameters alone; the effective one is a property of every other borrower in the same market and of the venue they will all be sold into. Two positions on identical terms, in two markets with identical parameters, do not have the same attachment point — the one whose neighbours' liquidation prices are more concentrated has the worse one, and nothing either borrower signed records the difference.

6 One inequality, three failures

The three ways a market of this type stops behaving as a tranche structure are not independent phenomena. They are (3) failing — on two terms, one of which fails in two qualitatively different ways.

term	**failure**	**observed as**
σ undefined	no price exists	liquidation and collateral withdrawal both impossible; borrowing halted; only un-borrowed liquidity remains withdrawable
$\sigma > \sigma^*$	collateral unsellable	bad debt on gapping or illiquid assets
$A/\text{LIF} < r_{\min}$	position too small	dust nobody clears; loss recognised late or never

The first row is not a limiting case of the second. An oracle that reverts does not make the collateral expensive to sell; it makes the solvency predicate (1) unevaluable, so the transaction fails before reaching any venue. Note also what it does *not* freeze: lenders whose funds are not currently borrowed can still withdraw, and typically do, which raises utilisation on the remainder and — by the interest-rate model — the rate charged to borrowers who cannot withdraw collateral. The freeze is asymmetric.

The third row is stated on collateral rather than on debt, since a liquidator can never repay more than A/LIF; when even that maximum repayment is below $r_{\min}$ no profitable liquidation exists at any size, and "never cleared" follows without hedging. It deserves emphasis because it inverts a common intuition. Large positions are not the dangerous ones *on this term*: they are the ones the gas condition is certain to clear — though, by Proposition 15, they may well fail the other. The two size conditions run in opposite directions, $r_{\min}$ from below and $N^* q_{\max}$ from above, so a market is enforceable only on the interval between them, and a sufficiently thin venue closes that interval entirely.

A protocol that records loss only at exhaustive seizure — as Morpho Blue does, writing bad debt when and only when the remaining collateral reaches zero — therefore does not recognise the loss on precisely the positions that fail this term. The accounting is correct at every step and lags reality for as long as $A/\text{LIF} < r_{\min}$ holds, which, absent

intervention, is forever.

7 What the cap asserts

The loan-to-value cap is read as a leverage limit. Proposition 10 gives it a second reading, and unlike the first it is quantitative.

Since $\sigma^* = c(1-\lambda)$, raising the cap thins the residual tranche and narrows the enforcing agent's slippage budget in strict proportion, at thirty basis points of budget per point of cap. At $\lambda = 0.98$ the market clears only if the collateral can be sold within sixty basis points of its oracle price; at $\lambda = 0.86$ the budget is 4.2 points. Neither number is a statement about volatility, and neither is enforced anywhere in the protocol — σ^* appears in no contract, and nothing reverts when it is exceeded. What happens instead is that nobody sends the transaction.

Setting a cap at λ is therefore an assertion — untested when made, and tested only in the worst state — that the collateral will trade within $c(1-\lambda)$ of its oracle price at the size the market has accumulated. Read that way, a high cap is not a permission granted to borrowers but a claim made on their behalf about a venue, and a low cap is better read as a disclosure about the collateral than as a control on the borrower.

8 What this does not settle

The model treats σ as a primitive and the collateral price as exogenous to execution beyond the impact terms. Four things are parametrised rather than derived, and each is a distinct piece of work.

Impact persistence and recovery. Propositions 15 and 17 carry ρ and Δ as constants. Both are properties of the venue and of the information content of the flow — a liquidation is uninformed selling, which should recover faster than informed — and both are estimable per venue. Deriving them requires a market microstructure model this paper does not supply, and N^* is only as good as ρ.

The shape of σ on a concentrated-liquidity venue. The results use only that σ is increasing; the numerical illustrations would change with the depth profile. On a venue where liquidity is placed in ranges, σ is not smooth and $q_{\max}$ can move discontinuously as the price crosses a range boundary — which is precisely the region where the frontier is tested. This deserves separate treatment.

Competition among liquidators. $\pi > 0$ is necessary for one agent to act; the realised clearing rate is set by the fastest, and priority auctions redistribute the premium without moving the frontier. Whether competition shortens Δ enough to matter is an empirical question about block construction rather than about lending.

Endogeneity of r_b. The rate responds to utilisation, and utilisation responds to whether the market clears. Proposition 17 takes r_b as given; the fixed point of rate, utilisation and exit behaviour is the natural object and remains open.

None of these weakens the frontier itself. (3) is a necessary condition, so a market failing it fails to clear regardless of what the extensions add. The conditions derived from it run the other way: Propositions 15 and 20 give sufficient conditions for survival and for the location of the stopping point, and a market satisfying them may still fail on a term

this model holds fixed.

References

[1] R. Cifuentes, G. Ferrucci and H. S. Shin. Liquidity risk and contagion. *Journal of the European Economic Association*, 3(2–3):556–566, 2005.

[2] F. Duarte and T. M. Eisenbach. Fire-sale spillovers and systemic risk. *Journal of Finance*, 76(3):1251–1294, 2021.

[3] L. Eisenberg and T. H. Noe. Systemic risk in financial systems. *Management Science*, 47(2):236–249, 2001.

[4] R. Greenwood, A. Landier and D. Thesmar. Vulnerable banks. *Journal of Financial Economics*, 115(3):471–485, 2015.

[5] A. Tarski. A lattice-theoretical fixpoint theorem and its applications. *Pacific Journal of Mathematics*, 5(2):285–309, 1955.